

PRATIK NANASO RAUT

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PROFESSIONAL SUMMARY

MCA graduate with strong proficiency in C++ programming, object-oriented design, data structures, and STL. Built production-quality systems projects including a virtual file system with inode-based memory management, a custom relational SQL engine using C++ STL containers, and a multi-threaded network server using POSIX threads. Experienced in writing clean, maintainable code, debugging complex issues, and working in Agile development environments with Git version control.

TECHNICAL SKILLS

C/C++: C, C++ (OOP, STL, Templates), Data Structures, Algorithms, Software Design Patterns

Multithreading & Networking: POSIX Threads, Mutex, Semaphores, TCP/IP Socket Programming, Concurrency

Systems & OS: Memory Management, /proc Filesystem, Inode Management, File Locking, Linux Internals

Other Languages: Java, Python, SQL, JavaScript, Bash/Shell Scripting

Tools & Environment: Git, GitHub, VS Code, GCC/G++, GDB Debugger, Maven, Linux (Ubuntu), Agile/Scrum

PROFESSIONAL EXPERIENCE

SetTribe IT Solutions

Feb 2026 – Jul 2026

Software Developer Intern

Pune, Maharashtra

- Developed and debugged application modules using Java and Spring Boot within Agile sprint cycles with daily standups.
- Wrote clean, maintainable code following OOP design patterns and participated in code reviews and software development activities.
- Troubleshooted and resolved software issues in backend services, improving module reliability by 40%.
- Managed Git-based version control workflows including branching, merging, and collaborative development.

PROJECTS

Customised Virtual File System (CVFS) | C++, OOP, Data Structures, Memory Management

- Simulated Linux file system operations (open, read, write, lseek) in C++ using inode-based metadata management and a custom interactive shell interpreter.
- Achieved O(1) file lookup by designing hash table-based inode indexing with chaining for collision resolution.
- Implemented in-memory file allocation table using C++ classes managing file permissions, ownership flags, and hierarchical directory traversal.
- Applied OOP principles (encapsulation, inheritance, polymorphism) for clean, modular, and maintainable code architecture.

Customized DBMS – Relational SQL Engine | C++, OOP, STL Containers, Data Structures

- Designed SQL query parser in C++ using OOP class hierarchy, supporting full DDL (CREATE, DROP) and DML (INSERT, SELECT, UPDATE, DELETE) operations.
- Engineered B-tree indexing structure for accelerated record lookup, reducing sequential scan overhead on large in-memory datasets.
- Simulated multi-table relational schema with join operations, constraint enforcement, and memory-managed record storage using C++ STL containers (vector, map, unordered_map).
- Implemented robust error handling and debugging routines for query validation and edge-case management.

Concurrent FTP Server – Multi-Client File Transfer | C, POSIX Threads, Multithreading, TCP/IP Sockets

- Engineered concurrent client handling using POSIX threads over TCP sockets, supporting simultaneous file transfers without data corruption.
- Implemented thread-safe resource management using mutex and semaphore primitives, eliminating race conditions under concurrent load.
- Designed application-layer protocol over raw sockets handling authentication, file listing, upload, and download command sequences.
- Debugged and resolved complex concurrency issues including deadlocks, resource contention, and buffer overflow scenarios.

EDUCATION

Dnyansagar Institute of Management and Research, Pune (SPPU)

2024 – 2026

Master of Computer Applications (MCA) – CGPA: 7.32

Pune, Maharashtra

C.T. Bora College, Shirur (SPPU)

2021 – 2024

Bachelor of Computer Applications (BCA) – Marks: 73.34%

Pune, Maharashtra